Question: A Mars rover sends data packets to Earth. Due to space interference, only 80 out of 100 packets are successfully received. If NASA randomly selects one received packet for analysis, what is the probability that it was successfully transmitted?

$$\text{Probability of an event occurring} = \frac{\text{no. of outcomes associated with the event}}{\text{Total number of outcomes}}$$

$$= 80 / 100$$

$$= 0.8$$

Question: A space station has 50 maintenance robots. Of these, 10 have recently reported minor system errors. If the station's AI selects one robot at random for inspection, what is the probability that the chosen robot has no system error?

$$\text{Probability of an event occurring} = \frac{\text{no. of outcomes associated with the event}}{\text{Total number of outcomes}}$$

$$= (50 – 10) / 50$$

$$= 0.8$$

Question: In a self-driving car, the sensor detects obstacles with a 90% accuracy when an obstacle is actually present and incorrectly detects an obstacle 10% of the time when no obstacle is present. In a test, the car encounters 200 situations with obstacles and 100 without. Out of the 200 situations with obstacles, the sensor correctly detects 180 obstacles, and out of the 100 situations without obstacles, the sensor incorrectly detects 15 obstacles. Given that the sensor detects an obstacle, what is the probability that an obstacle is actually present?

$$P(\text{Actual Obstacle}|\text{Detection}) = \frac{\text{Number of correct detections}}{\text{Total number of detections}}$$

$$= 180 / 195 = 12 / 13 \approx 0.923$$

Question: In a cutting-edge quantum computing lab, an advanced system performs auto-error correction to ensure accurate computations. The probability that a computation is error-free, given that the system applied auto-error correction, is 0.95. Additionally, the system engages in auto-error correction for 70% of computations. If a scientist randomly inspects a computation, what is the probability that it was both error-free and had auto-error correction applied?

$$\text{Probability of event A happening given B} = \frac{\text{Probability of both events A and B happening}}{\text{Probability of event B happening}}$$

Probability of both events A and B happening = 0.95 * 0.7 = 0.665

A school administers a math test to 100 students. The scores of 99 students range from 70 to 90, with most of them clustered around 80. However, the top-performing student (who is a prodigy) scores 150.

The school is advertising its performance in the test to attract more students and make itself appear as a high-achieving institution. It is considering three different metrics for advertising its test scores:

- Mean Score

- Median Score
- Mode Score

Which metric should the school advertise to make its performance appear as impressive as possible, and why?

Mean = ((99 * 80) + (1 * 150)) / 100 = 8070 / 100 = 80.7

Median = 80 (50th and 51st score)

Mode = 80